

Stephen A. Rice

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HIGHLIGHTS

- Master's in Computer Science with a focus on systems and machine learning
- 4 years of professional development experience with fluency in Rust, Go, C#, Python, and Java
- Built and maintained backend systems, APIs, and low latency data pipelines for IoT, desktop, and web applications
- Experience with containerized deployments (Docker, Kubernetes) and cloud platforms (AWS, GCP)
- Strong grasp of software practices: testing, CI/CD, version control, agile
- Clear communicator, reliable collaborator, and strong sense of ownership

PROFESSIONAL SUMMARY

Software engineer with four years of experience building reliable backend systems in Rust, Go, and C#. I've led small projects end-to-end and contributed to production software across IoT, desktop, and web applications. With a background in engineering, data analysis, and information security, I bring strong analytical instincts, practical programming skill, and a steady drive to learn whatever a project requires.

WORK EXPERIENCE

Software Engineer - Promess Incorporated, Brighton, MI
February 2021 to May 2025

Project Lead Contributions

- MotionPro/UltraPro Connectors — C#/.NET, Rust, gRPC, Linux
 - Designed and implemented API servers that interfaced with industrial press controllers, providing stable, well-defined interfaces used across new customer deployments.
 - Took ownership of the project lifecycle, delivering a reliable system with minimal defects and enabling new application integrations.
- BDD and TDD Tools — Rust
 - Built automated behavior-monitoring and end-to-end testing tools that surfaced defects earlier in development, reducing release-time issues and improving team efficiency.

Team Contributions

- UltraPro — C#/.NET, C++, XML
 - Tested, debugged, and maintained production software and embedded firmware for the company's flagship press control system.
 - Improved reliability and supported successful releases across wide customer usage.
- UltraPro2 — Rust, GraphQL, WASM, Linux
 - Contributed to the system architecture, R&D, and feature development for a modern rework of the UltraPro platform.
 - Helped design event-driven components for real-time data processing and improved system clarity for future development.
- Workflow Director — Go, MQTT
 - Built the backend for a barcode-scanning microservice that integrated with UltraPro systems.
 - Designed message-handling and device communication logic that made the tool reliable enough for production use and adoption by customers.
- Data Utility — SQL
 - Optimized data retrieval logic to reduce load times from several minutes to under a second, significantly improving responsiveness and usability for operators and engineers.

Additional Contributions

- Deployed IoT applications in containerized environments (Docker, Kubernetes) for consistent updates, orchestration, and easier long-term maintenance.
- Collaborated using agile practices and modern SDLC tools including Azure DevOps, GitHub, Jira, and Confluence.

Graduate Assistant - Eastern Michigan University, Ypsilanti, MI

January 2019 to December 2020

- Coded with multi-threaded web scraping and syntax analyzers to mine Python files on Github as a research assistant, to be used for NLP and machine learning
- Tutored in Java and Python
- Aided as a teaching assistant in applied cryptography

EDUCATION

M.S. in Computer Science - Eastern Michigan University

- Coursework: algorithms, data structures, networks, embedded systems, parallel processing (CUDA), functional programming, machine learning, AI
- Thesis: trained a large language model for punctuation prediction using PyTorch and GCP
- Experience leading teams as Scrum Master

B.S. in Nuclear Engineering & Radiological Sciences - University of Michigan

- Focused on energy systems, risk analysis, and computational modeling
- Programming experience in C++, Python, and Matlab

SKILLS

Languages

Rust, Go, C#, Python, Java, C++, SQL, JavaScript, TypeScript, Matlab

Frameworks & APIs

REST APIs, gRPC, GraphQL, protobuf, Swagger/OpenAPI, WebAssembly (WASM), MQTT, Node.js, NPM

Tools & Platforms

Docker, Kubernetes, Git, CI/CD, Azure DevOps, Visual Studio, WSL

Cloud

Amazon Web Services (AWS), Google Cloud Platform (GCP)

Systems & Paradigms

Embedded software, Linux development, multithreaded/asynchronous programming, functional programming, streaming data pipelines, object-oriented design, networking fundamentals

AI / Data

Machine learning & AI basics, data visualization, data analysis

Practices

Testing & debugging, quality assurance, modeling, research, documentation, agile development (Scrum)